NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
ELECTRONIC DEVICES AND CIRCUITS (MVPC 201)
Class Test-1st (2025-2026)

Time: 50 min. **MM. 15**

NOTE: Attempt three questions. Assume suitable data, if required.

Q. 1 Derive the expression of the diode current. Draw the I-V characteristics from the derived equation.

Q. 2 Sketch the output voltage shown in Fig. 2 (Consider Silicon diodes) in Fig 1.

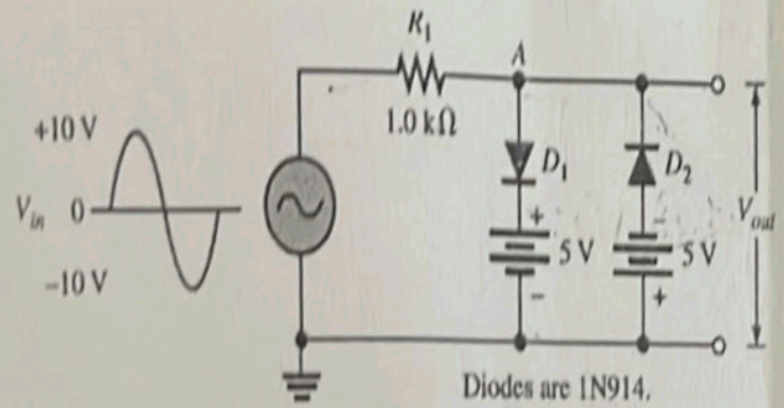

Fig. 1

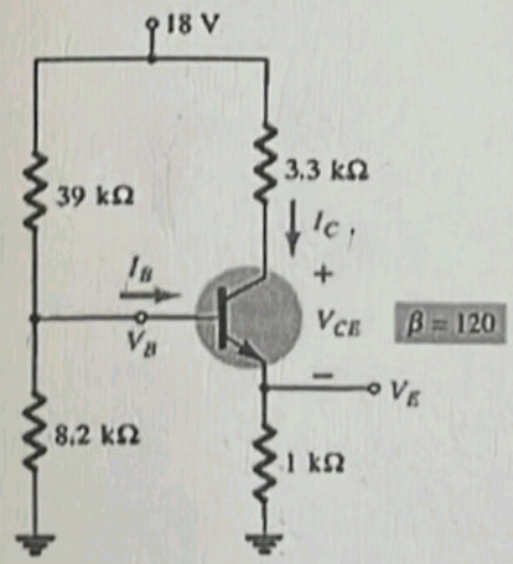

Fig. 2

Q. 3 Determine the following for the voltage-divider configuration of Fig. 2.

a) I_C . b) V_{CE} . c) I_B . d) V_E . e) V_B

Q. 4 Differentiate between a rectifying and an ohmic contact. Consider a Metal-Semiconductor (n-type) junction and explain using an energy band diagram.